

Zero Trust Architecture Simulation

Validating secure access through identity, device, and contextual verification to protect organizational assets.



Core Security Pillars

A comprehensive model for verifying access at every layer of the infrastructure.

| User Identity & Personas

Internal Roles

HR & Finance: Access to sensitive employee records and financial databases with elevated verification.

Admin: Privileged access to the console and infrastructure backend using robust MFA.

External & Guests

Partners/Guests: Restricted access to specific public portals and collaborative tools.

Identity Signals: Verification based on role, location, and behavioral consistency.

| Device Health Integrity



Managed

Corporate-owned assets with centralized policy control and visibility.



Compliant

Verifying patch levels, encryption, and active security agents are present.



Risk Score

Dynamic assessment of device health and historical behavioral anomalies.

Contextual Access Requests

Unified Request Logic

Every access request is evaluated by combining three critical data points:

- ✓ **Identity:** Verified User Credentials
- ✓ **Device:** Verified Hardware Health
- ✓ **Segment:** Specific Access Path Path

Implicit trust is eliminated; authorization is session-specific.



| Verification Strategy

100%

EXPLICIT VERIFICATION

Policy Decision Engine

The "Brain" of the simulation. It receives the access request and consults organizational policies to "Verify Explicitly."

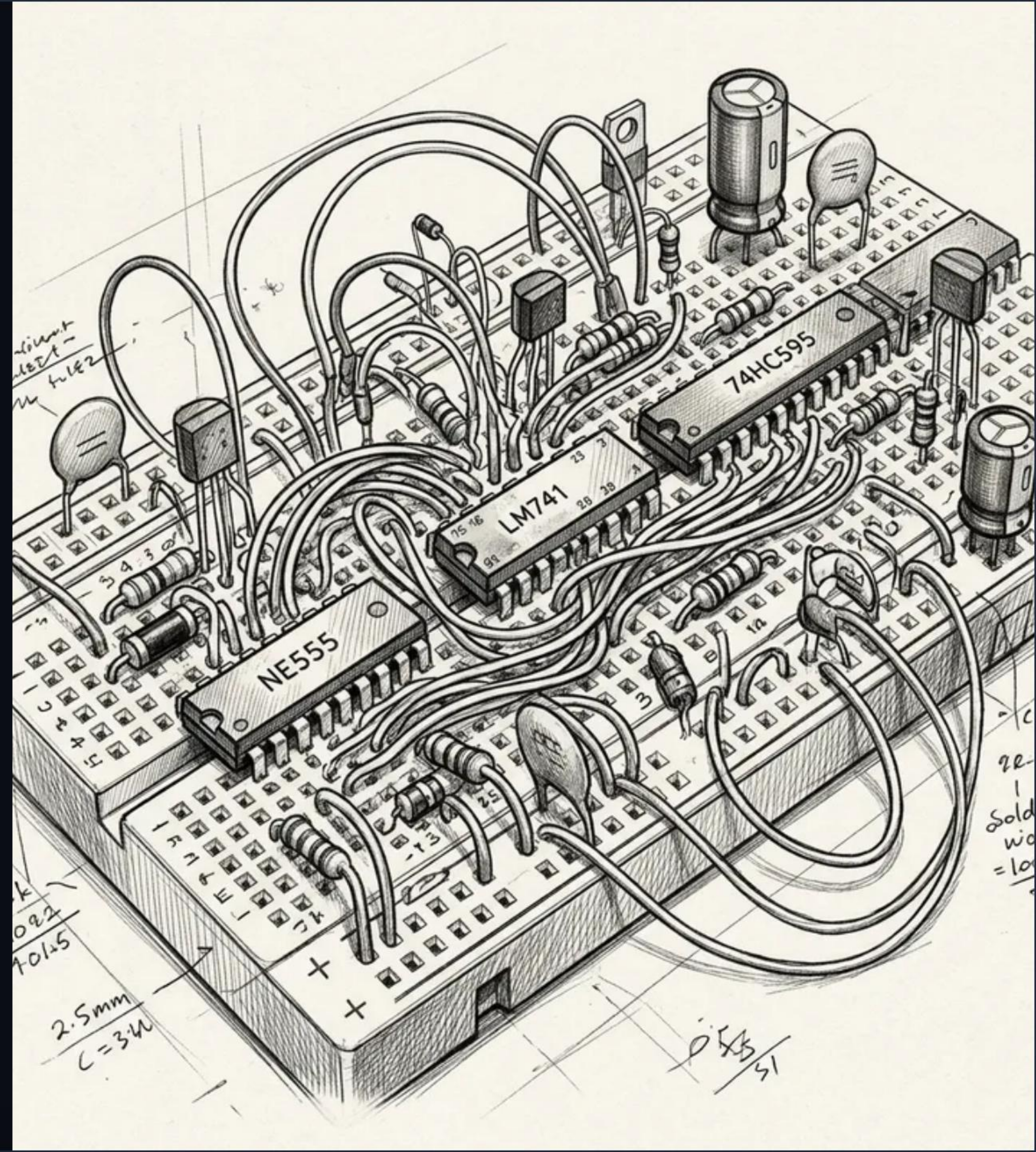
No request is allowed until all security hurdles—Identity, Device Posture, and Micro-segmentation rules—are cleared.

The PEP Gatekeeper

Policy Enforcement Point

The Policy Enforcement Point (PEP) acts as the final bouncer. It intercepts every request and executes the ruling from the Decision Engine.

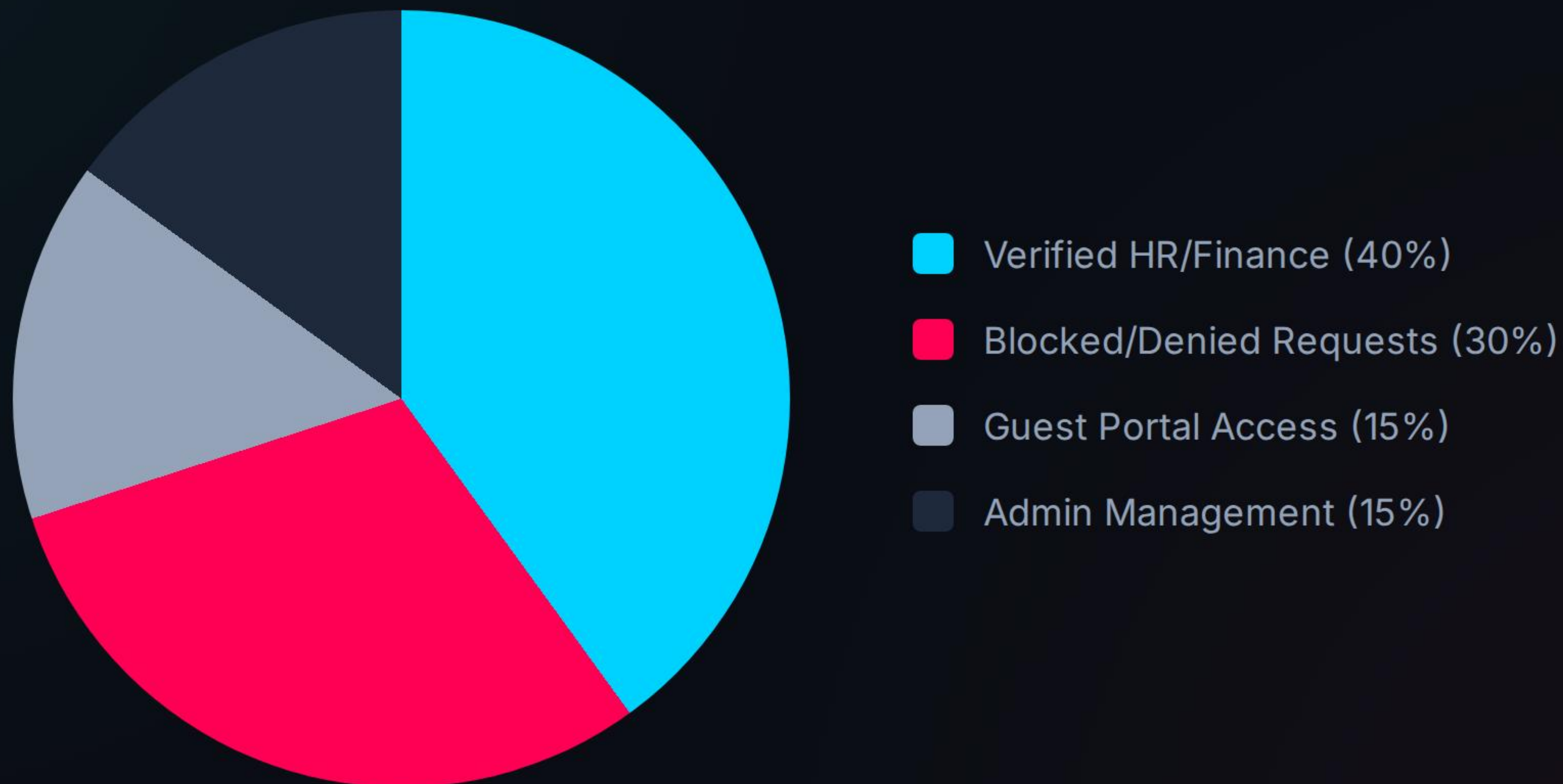
Actions are binary: **Allow** or **Deny**. This ensures that unauthorized actors are blocked at the perimeter of the specific resource zone.



Protected Resource Zones

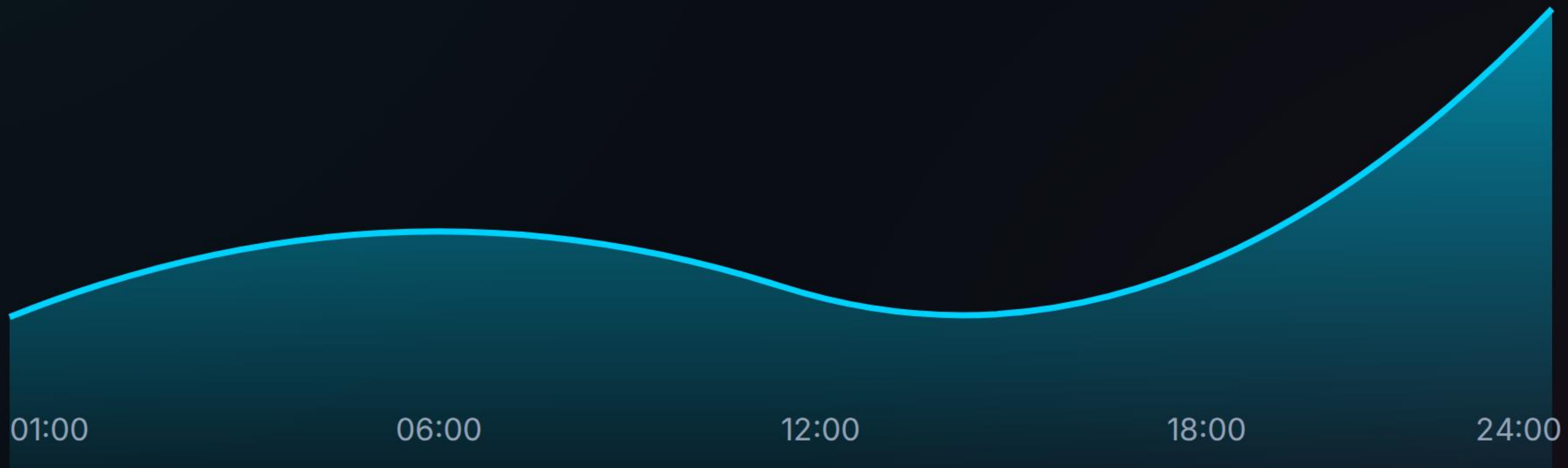
Asset Name	Classification	Primary User Role	Access Policy
HR Records	PII / Sensitive	HR Team Only	Least Privilege
Finance Database	Confidential	Finance Analysts	Role-Based (RBAC)
Admin Console	Critical	IT System Admins	Just-in-Time (JIT)
Public Portal	External	Guests / Partners	Public Path Only

Simulated Traffic Distribution



The simulation highlights a significant volume of denied traffic (30%), indicating effective policy enforcement against unauthorized attempts.

| Security Monitoring Trends



The Monitoring Engine tracks "Alerts + Repeated Denials" in real-time, allowing security teams to identify potential lateral movement attempts or credential stuffing attacks.

Forensic Evidence & Visibility



Detailed CSV Logs

Historical records of every decision made by the Policy Engine for auditing.



Packet Capture

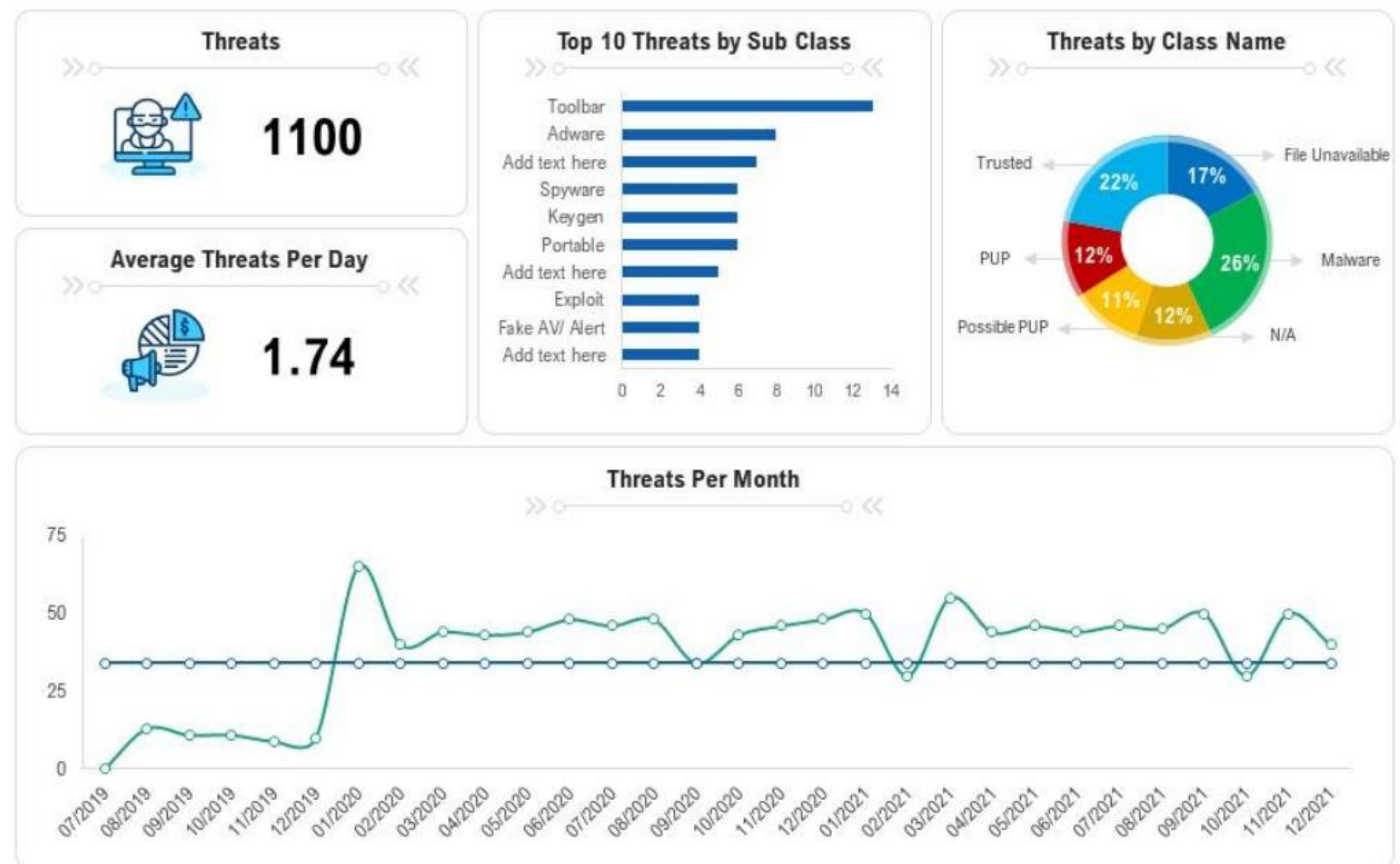
Deep inspection of traffic flows to validate segment-to-segment communication.

Real-time Alerts

Immediate notification for policy violations and unauthorized device detection.

Cyber Security Dashboard for Security Operations

This slide Covers the Graphical Representation Analysis of Cyber Security. Data covered in this dashboard includes-Threats, Threats per month, Average threat per day with top 10 threat by sub class and class name.



This graph/chart is linked to excel, and changes automatically based on data. Just left click on it and select "edit data".

Questions?

Ensuring your zero trust transformation is data-driven and secure.

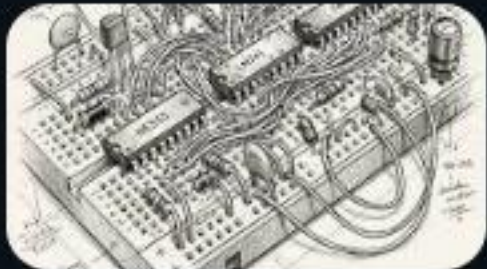
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| Image Sources



https://media.istockphoto.com/id/1442836782/vector/passwordless-login-with-passkey-concept-3d-illustration.jpg?s=612x612&w=0&k=20&c=-HcCi_o64BUWKQTMC4jUbDSuMnDGWQK5ctrKzgxlck=

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Source: easy-peasy.ai



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